

Lab 6: Joins and Cross Products

Instructions

- 1. You must use cross products to solve the first 5 questions (Do not use inner/outer joins). For the remaining questions (Q6-Q10), use inner or outer joins (Do not use cross joins). You are not allowed to use subqueries or set operations for any of the questions.
- 2. If your output matches the expected output, don't worry about the ordering of rows in your result.

Question 1

List each student’s name along with the department name of their major sorted by studentName in ascending order.

Expected output:

studentName	deptName
Aarav	Computer Science
Aditi	Electrical
Bhavya	Computer Science
Bilal	Civil
Chetan	Computer Science
Chirag	Computer Science
Dev	Mechanical
Diya	Computer Science
Eshan	Computer Science
Farah	Computer Science
Gaurav	Mechanical
Hina	Mechanical
Ishaan	Mechanical
Juhi	Mechanical
Karan	Mechanical
Lata	Mechanical
Manav	Electrical
Nisha	Electrical
Om	Electrical
Pooja	Electrical
Qadir	Electrical
Riya	Electrical
Sahil	Civil
Tina	Civil
Uday	Civil
Vani	Civil
Wasim	Civil
Xena	Civil
Yash	Computer Science
Zara	Mechanical

30 rows in set (0.000 sec)

Question 2

For each course offering (course, semester, year), display the course name, semester, year, and the total number of students enrolled.

Expected output:

courseName	semester	year	total_students
AI	Fall	2024	3
Algebra	Fall	2023	2
Calculus	Spring	2024	2
Circuits	Fall	2023	3
Control	Spring	2023	2
DBMS	Fall	2023	3
DBMS	Fall	2024	1
DBMS	Spring	2024	1
Design	Spring	2024	4
Dynamics	Fall	2024	3
GeoTech	Spring	2024	3
Hydrology	Spring	2023	3
Networks	Spring	2023	2
OS	Fall	2024	3
Power	Fall	2024	4
Robotics	Spring	2023	3
Signals	Spring	2024	4
Structures	Fall	2023	2
Thermo	Fall	2023	3
Transport	Fall	2024	4

20 rows in set (0.007 sec)

Question 3

List instructors (full name) who are teaching courses outside their own department.

Expected output:

instructorId	firstName	lastName
101	Alan	Smith

1 row in set (0.000 sec)

Question 4

List pairs of students who belong to the same department but are in different years, with the constraint that at least one of their names should start with 'A'. Do not list duplicate or reversed pairs.

Expected output:

studentName	studentName
Aarav	Bhavya
Aarav	Chetan
Aarav	Diya
Aarav	Eshan
Aarav	Farah
Aarav	Chirag
Manav	Aditi
Nisha	Aditi
Pooja	Aditi
Qadir	Aditi

10 rows in set (0.002 sec)

Question 5

For each department, display the department name and the total number of distinct students enrolled in its courses. Do not include departments with no students enrolled.

Expected output:

deptName	total_students
Civil	6
Computer Science	6
Electrical	6
Mathematics	2
Mechanical	6

5 rows in set (0.001 sec)

Question 6

List all students along with the number of sections they are enrolled in, sorted by studentId in ascending order. Include students not enrolled in any section.

Expected output:

studentId	studentName	total_sections
201	Aarav	3
202	Bhavya	2
203	Chetan	2
204	Diya	2
205	Eshan	2
206	Farah	2
207	Gaurav	3
208	Hina	2
209	Ishaan	2
210	Juhi	2
211	Karan	2
212	Lata	2
213	Manav	3
214	Nisha	2
215	Om	2
216	Pooja	2
217	Qadir	2
218	Riya	2
219	Sahil	2
220	Tina	2
221	Uday	2
222	Vani	2
223	Wasim	2
224	Xena	2
225	Yash	2
226	Zara	2
227	Aditi	0
228	Bilal	0
229	Chirag	0
230	Dev	0

30 rows in set (0.000 sec)

Question 7

List all sections taught in the 'Spring' semester, along with the names of students enrolled in them, sorted by sectionId in descending order. Include sections that have no enrollments.

Expected output:

sectionId	studentName
325	NULL
323	NULL
321	NULL
319	Yash
319	Zara
317	Uday
317	Vani
317	Xena
315	Sahil
315	Tina
315	Wasim
313	Om
313	Pooja
311	Manav
311	Nisha
311	Qadir
311	Riya
309	Hina
309	Karan
309	Lata
307	Gaurav
307	Ishaan
307	Juhi
307	Lata
304	Chetan
304	Eshan
302	Chetan

27 rows in set (0.000 sec)

Question 8

Find instructors who are not teaching any section.

Expected output:

instructorId	firstName	lastName
110	Jack	White

1 row in set (0.000 sec)

Question 9

Find departments that offer courses but have no students enrolled in any of their sections.

Expected output:

deptId	deptName
6	Bioinformatics

1 row in set (0.001 sec)

Question 10

For each instructor, display their full name and the total stipend paid to lab assistants working in sections taught by them. Include instructors who do not teach any section or whose sections have no lab assistants.

Expected output:

instructorId	firstName	lastName	total_stipend
101	Alan	Smith	52500.00
102	Brian	Smith	NULL
103	Clara	Jones	9000.00
104	David	Miller	9000.00
105	Eva	Brown	NULL
106	Frank	Taylor	29200.00
107	Grace	Taylor	NULL
108	Henry	Moore	9200.00
109	Irene	Moore	NULL
110	Jack	White	NULL
111	Kelly	Wilson	NULL
112	Liam	Wilson	25500.00
113	Mona	Clark	23300.00
114	Nate	Clark	9400.00
115	Omar	Hall	NULL
116	Paul	Allen	40400.00
121	Una	Scott	20500.00

17 rows in set (0.001 sec)

◀ Lab 6 zip

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Lab 6 solution upload link (.sql file) ▶